

QuantMS nextflow workflow (nf-core compatible)

23th February 2026

Henry Webel

Does broccoli boost bad gut bacteria?

- One strain of *E. coli* was analyzed using MS-based proteomics
- Shoutout to [Caroline Jachmann](#) for creating a peptide atlas for *E. coli* and sharing information on small scale experiments

<https://www.hudson.org.au/news/does-broccoli-boost-bad-gut-bacteria/>

INFLAMMATION

NEWS | INFLAMMATION | MICROBIOME IN HEALTH AND DISEASE

Does broccoli boost bad gut bacteria?

22 August 2023

By Rob Clancy, staff writer. Reviewed by Dr Emily Gulliver

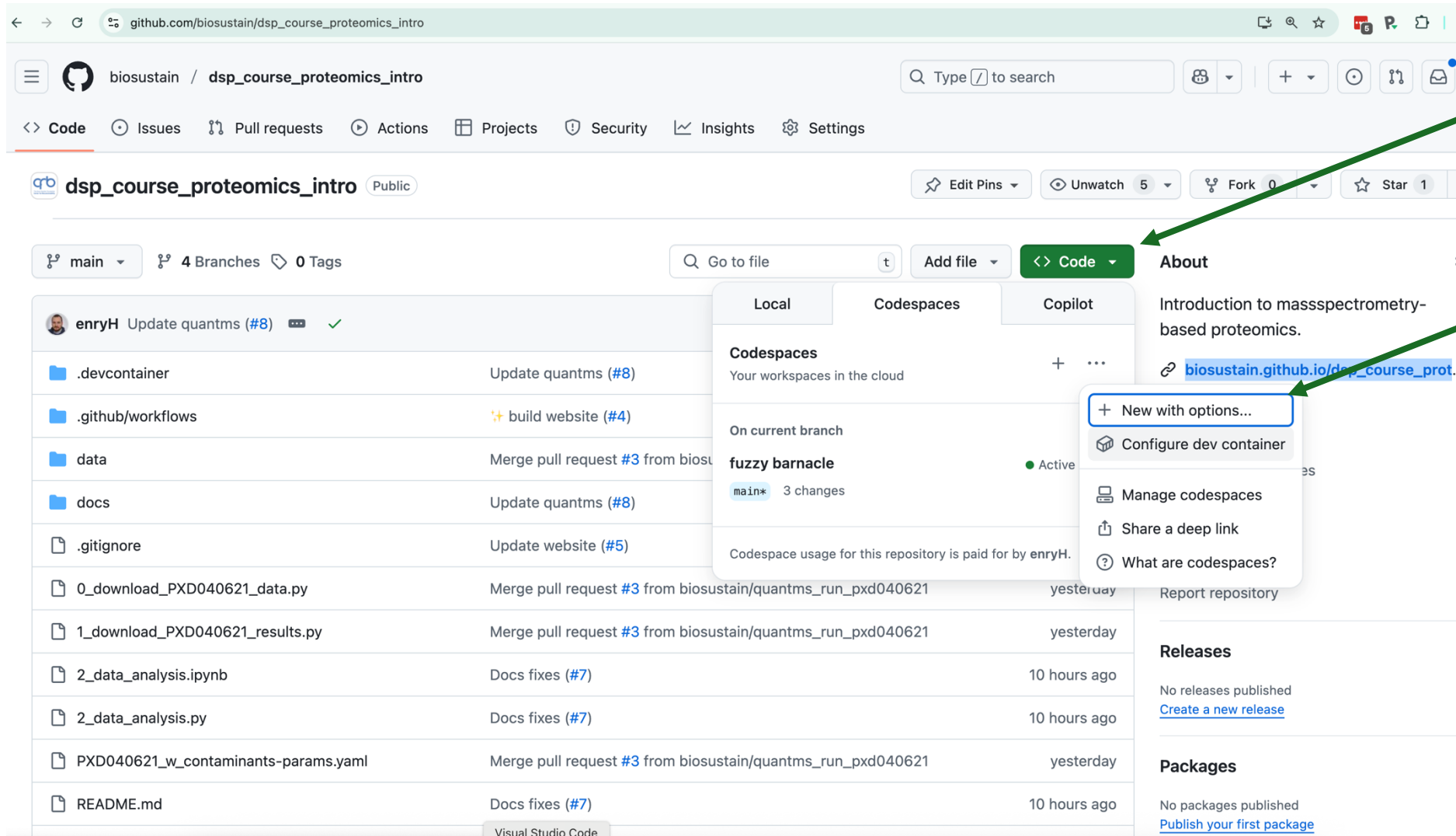


Dr Emily Gulliver

Latest research into the human microbiome begins to untangle how broccoli can alter healthy gut bacteria.

Cruciferous vegetables, including broccoli, cauliflower, brussels sprouts and kale, are often recommended due to the presence of the antioxidant sulforaphane, which is thought to be beneficial for general health and wellbeing and in treating diseases such as cancer.

Let's start the workflow first



github.com/biosustain/dsp_course_proteomics_intro

biosustain / dsp_course_proteomics_intro

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dsp_course_proteomics_intro Public

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main 4 Branches 0 Tags

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Local Codespaces Copilot

Codespaces
Your workspaces in the cloud

On current branch

fuzzy barnacle
main* 3 changes

Codespace usage for this repository is paid for by enryH.

+ New with options...
Configure dev container
Manage codespaces
Share a deep link
What are codespaces?

enryH Update quantms (#8) ✓

.devcontainer	Update quantms (#8)
.github/workflows	build website (#4)
data	Merge pull request #3 from biosustain/quantms_run_pxd040621
docs	Update quantms (#8)
.gitignore	Update website (#5)
0_download_PXD040621_data.py	Merge pull request #3 from biosustain/quantms_run_pxd040621 yesterday
1_download_PXD040621_results.py	Merge pull request #3 from biosustain/quantms_run_pxd040621 yesterday
2_data_analysis.ipynb	Docs fixes (#7) 10 hours ago
2_data_analysis.py	Docs fixes (#7) 10 hours ago
PXD040621_w_contaminants-params.yaml	Merge pull request #3 from biosustain/quantms_run_pxd040621 yesterday
README.md	Docs fixes (#7) 10 hours ago

Visual Studio Code

Introduction to massspectrometry-based proteomics.
[biosustain.github.io/dsp_course_prot...](#)

Releases
No releases published
[Create a new release](#)

Packages
No packages published
[Publish your first package](#)

Codespace with 4 cores (and 16GB of memory)

Create codespace for **biosustain/dsp_course_proteomics_intro**

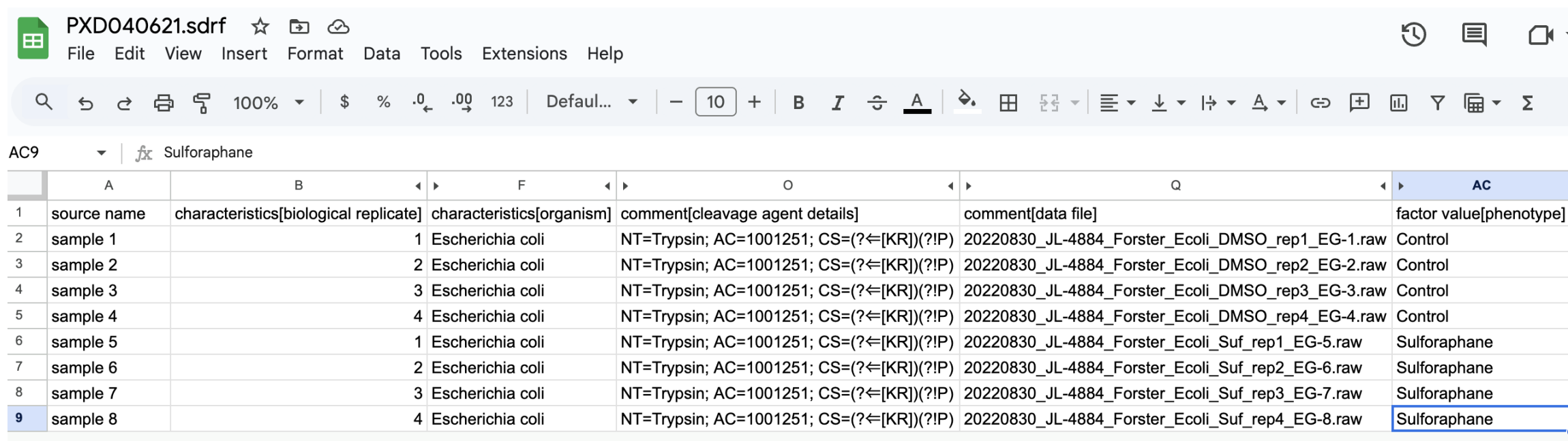
Branch This branch will be checked out on creation	 main ▾
Dev container configuration Your codespace will use this configuration	nextflow-training ▾
Region Your codespace will run in the selected region	Europe West ▾
Machine type Resources for your codespace	4-core ▾
Create codespace	

1

2

Sample Data Relationship Format (SDRF) file

Describes the design of an experiment (metadata): Defines in a standardized manner details about how the experiment was performed - sample information, organism, phenotype/grouping, etc.



	A	B	F	O	Q	AC
1	source name	characteristics[biological replicate]	characteristics[organism]	comment[cleavage agent details]	comment[data file]	factor value[phenotype]
2	sample 1		1 Escherichia coli	NT=Trypsin; AC=1001251; CS=(?<[KR])(?!P)	20220830_JL-4884_Forster_Ecoli_DMSO_rep1_EG-1.raw	Control
3	sample 2		2 Escherichia coli	NT=Trypsin; AC=1001251; CS=(?<[KR])(?!P)	20220830_JL-4884_Forster_Ecoli_DMSO_rep2_EG-2.raw	Control
4	sample 3		3 Escherichia coli	NT=Trypsin; AC=1001251; CS=(?<[KR])(?!P)	20220830_JL-4884_Forster_Ecoli_DMSO_rep3_EG-3.raw	Control
5	sample 4		4 Escherichia coli	NT=Trypsin; AC=1001251; CS=(?<[KR])(?!P)	20220830_JL-4884_Forster_Ecoli_DMSO_rep4_EG-4.raw	Control
6	sample 5		1 Escherichia coli	NT=Trypsin; AC=1001251; CS=(?<[KR])(?!P)	20220830_JL-4884_Forster_Ecoli_Suf_rep1_EG-5.raw	Sulforaphane
7	sample 6		2 Escherichia coli	NT=Trypsin; AC=1001251; CS=(?<[KR])(?!P)	20220830_JL-4884_Forster_Ecoli_Suf_rep2_EG-6.raw	Sulforaphane
8	sample 7		3 Escherichia coli	NT=Trypsin; AC=1001251; CS=(?<[KR])(?!P)	20220830_JL-4884_Forster_Ecoli_Suf_rep3_EG-7.raw	Sulforaphane
9	sample 8		4 Escherichia coli	NT=Trypsin; AC=1001251; CS=(?<[KR])(?!P)	20220830_JL-4884_Forster_Ecoli_Suf_rep4_EG-8.raw	Sulforaphane

LessSDRF
tool helps to
create these
tables

[View online](#)

INPUTS

FASTA File with UNIPROT protein sequences

...

>sp|PODSF9|EVGL_ECOLI Protein EvgL OS=Escherichia coli (strain K12) OX=83333 GN=evgL PE=1 SV=1

MLHCKGNNL

>sp|PODSH1|YSAE_ECOLI Protein YsaE OS=Escherichia coli (strain K12) OX=83333 GN=ysaE PE=1 SV=1

MRNAVSKAGIISRRRLLLFQFAG

>sp|PODV20|YTCB_ECOLI Protein YtcB OS=Escherichia coli (strain K12) OX=83333 GN=yticB PE=1 SV=1

MHLQLIKDNIHSVVICYT

...

>CON_ENSEMBL:ENSBTAP00000038329 (Bos taurus) 9 kDa protein

LPENVTPPEQHKGTSVIHKAVLDVGEEGTEGA AVTAVVMATSSLLHTLTVSFNRPFLLSI

FCKETQSIIFLGKVTNPKEA

...

- 4413 known *E. coli* proteins
- 246 known contaminants sequences

- ThermoFisher instruments:
 - raw files
- Open standard, text based:
 - mzML files
- In this tutorial we use mzML files directly to skip the spectra extraction as quantms run on mzML files

One MS1 spectrum (DDA)

```
<spectrum id="controllerType=0 controllerNumber=1 scan=6" index="5" defaultArrayLength="736">
  <cvParam cvRef="MS" accession="MS:1000511" value="1" name="ms level" />
  <cvParam cvRef="MS" accession="MS:1000579" value="" name="MS1 spectrum" />
  <cvParam cvRef="MS" accession="MS:1000130" value="" name="positive scan" />
  <cvParam cvRef="MS" accession="MS:1000285" value="15248891" name="total ion current" />
  <cvParam cvRef="MS" accession="MS:1000127" value="" name="centroid spectrum" />
  <cvParam cvRef="MS" accession="MS:1000504" value="401.923461914063" name="base peak m/z" unitAccession="MS:1000040" unitName="m/z" unitCvRef="MS" />
  <cvParam cvRef="MS" accession="MS:1000505" value="2704116.25" name="base peak intensity" unitAccession="MS:1000131" unitName="number of detector counts" unitCvRef="MS" />
  <cvParam cvRef="MS" accession="MS:1000528" value="375.884124755859" name="lowest observed m/z" unitAccession="MS:1000040" unitName="m/z" unitCvRef="MS" />
  <cvParam cvRef="MS" accession="MS:1000527" value="1665.40612792969" name="highest observed m/z" unitAccession="MS:1000040" unitName="m/z" unitCvRef="MS" />
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    <cvParam cvRef="MS" accession="MS:1000795" value="" name="no combination" />
    <scan instrumentConfigurationRef="IC1">
      <cvParam cvRef="MS" accession="MS:1000016" value="6.0323342" name="scan start time" unitAccession="U0:0000031" unitName="minute" unitCvRef="U0" />
      <cvParam cvRef="MS" accession="MS:1000512" value="FTMS + p NSI Full ms [375.0000-1800.0000]" name="filter string" />
      <cvParam cvRef="MS" accession="MS:1000927" value="50" name="ion injection time" unitAccession="U0:0000028" unitName="millisecond" unitCvRef="U0" />
      <scanWindowList count="1">
        <scanWindow>
          <cvParam cvRef="MS" accession="MS:1000501" value="375" name="scan window lower limit" unitAccession="MS:1000040" unitName="m/z" unitCvRef="MS" />
          <cvParam cvRef="MS" accession="MS:1000500" value="1800" name="scan window upper limit" unitAccession="MS:1000040" unitName="m/z" unitCvRef="MS" />
        </scanWindow>
      </scanWindowList>
    </scan>
  </scanList>
  <binaryDataArrayList count="2">
    <binaryDataArray encodedLength="3152">
      <cvParam cvRef="MS" accession="MS:1000514" value="" name="m/z array" unitAccession="MS:1000040" unitName="m/z" unitCvRef="MS" />
      <cvParam cvRef="MS" accession="MS:1000523" value="" name="64-bit float" />
      <cvParam cvRef="MS" accession="MS:1000574" value="" name="zlib compression" />
      <binary>eJwT03LYVXuawPHjVumjGA1lG5dTGtMi3MYUG32496hJOiNX3EvLcJlNsEDNckGWlYogXaIwiU3AsUWcZ2TJEnEm5Wimg48C466j15M12ZRgU1paCTPP+/399Xne9fceFk3TzIeylxna/0Vjck5ovM26h1oDLdI
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    <binaryDataArray encodedLength="4356">
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      <cvParam cvRef="MS" accession="MS:1000523" value="" name="64-bit float" />
      <cvParam cvRef="MS" accession="MS:1000574" value="" name="zlib compression" />
      <binary>eJwTWHlCt+kXvILFSI2MTJi5ZWlAaTTGdSl+Ekt3KVtypUVl+dhr5TG4qo7Jk8FPWm32pjC1kuxULeu0JuSVDpozK0pdKpj3P/PV+3vu+73nPec45zznvFQRBUrdyQxIEQWzvf6ppFBz63sIYceZq06jafJHbNGr
    </binaryDataArray>
  </binaryDataArrayList>
</spectrum>
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<cvParam cvRef="MS" accession="MS:1000285" value="15248891" name="total ion current" />
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<cvParam cvRef="MS" accession="MS:1000504" value="401.923461914063" name="base peak m/z" unitAccession="MS:1000040" unitName="m/z" unitCvRef="MS" />
<cvParam cvRef="MS" accession="MS:1000505" value="2704116.25" name="base peak intensity" unitAccession="MS:1000131" unitName="number of detector counts" unitCvRef="MS" />
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<cvParam cvRef="MS" accession="MS:1000527" value="1665.40612792969" name="highest observed m/z" unitAccession="MS:1000040" unitName="m/z" unitCvRef="MS" />
<scanList count="1">
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<cvParam cvRef="MS" accession="MS:1000523" value="" name="64-bit float" />
<cvParam cvRef="MS" accession="MS:1000574" value="" name="zlib compression" />
<binary> ... 93jb+B7DdOsY=</binary>
</binaryDataArray>
<binaryDataArray encodedLength="4356">
<cvParam cvRef="MS" accession="MS:1000515" value="" name="intensity array" unitAccession="MS:1000131" unitName="number of counts" unitCvRef="MS" />
<cvParam cvRef="MS" accession="MS:1000523" value="" name="64-bit float" />
<cvParam cvRef="MS" accession="MS:1000574" value="" name="zlib compression" />
<binary>eJwTWHlct+ ... yNfr3NnHXsNq2F/69YyX/gWrVuzh</binary>
</binaryDataArray>
</binaryDataArrayList>
</spectrum>
```

QuantMS

- Nextflow is a workflow executor (analysis steps combined in an execution graph)
 - Reproducible and scalable
- Nf-core is a collection of workflows maintained by nextflow and an open-source community



A global community effort to collect a curated set of open-source analysis pipelines built using Nextflow.

Pipelines

Browse the 130 pipelines that are currently available as part of nf-core.

Released 80

Under development 38

Archived 12

↓ Last release ▾

☰ ☰

proteinfamilies ✓

☆ 14

Generation and update of protein families

metagenomics protein-families proteomics

📦 1.1.0 released 4 days ago

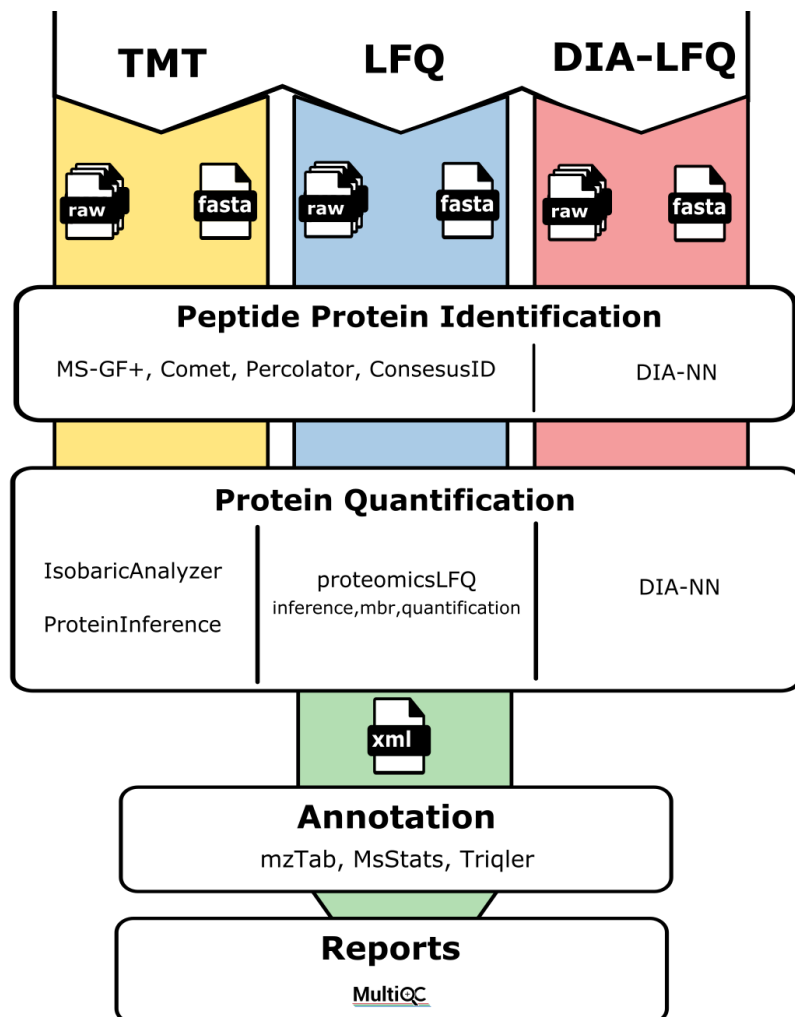
epitopeprediction ✓

☆ 44

A bioinformatics best-practice analysis pipeline for epitope prediction and annotation

epitope epitope-prediction mhc-binding-prediction

📦 3.0.0 released 5 days ago



Learn the Basics

Understand prerequisites and core concepts

→ [Getting started guide](#)

Step-by-Step

Follow detailed tutorials for your data type

→ [View tutorials](#)

Configure

Customize parameters for your analysis

→ [Parameter reference](#)

What type of data do you have?

DDA

Data-Dependent Acquisition

→ [DDA Analysis](#)

DIA

Data-Independent Acquisition

→ [DIA Analysis](#)

TMT/iTRAQ

Isobaric Labeling

→ [Isobaric Analysis](#)

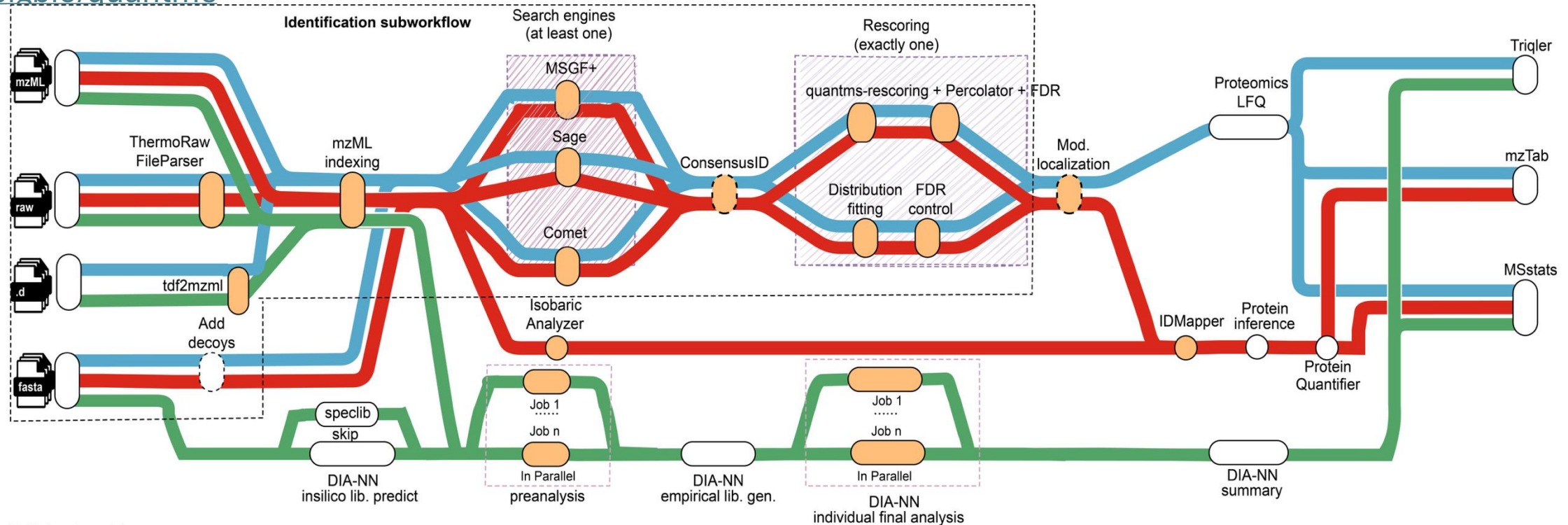
Label-Free

LFQ Quantification

→ [LFQ Analysis](#)

QuantMS (nf-core compatible)

bigbio/quantms



bigbio/quantms v1.3

Example analysis pathways

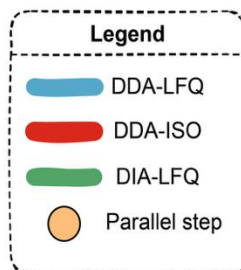


Table 8 Capabilities timeline

Capability	Introduced in	Notes
DDA-LFQ workflow	1.0.0	Label-free quantification; OpenMS-based quantification. See Release 1.0.0 .
Isobaric TMT workflow	1.2.0	TMT reporter quantification and MSstats(TMT) export. See Release 1.2.0 .
DIA workflow (DIA-NN based)	1.3.0	DIA identification/quantification and exports to MSstats and Triqler. See Release 1.3.0 .
Conda disabled by default	1.4.0	Container-first execution (Docker/Singularity/Podman). See Release 1.4.0 .
quantms-rescoring (DeepLC/MS2PIP integration)	1.5.0	Optional ML-based rescoring features for improved identifications. See Release 1.5.0 .
1.6.0 highlights	1.6.0	AlphaPeptDeep integration into quantms-rescoring. See Release 1.6.0 .
1.7.0	1.7.0	quantms-rescoring supports AlphaPeptDeep transfer learning. LuciPHORAdapter is replaced with onsite python package. See Release 1.7.0 .

See overview [here](#)

- Cloud native
- Containerized
- Focused on reproducibility and scalability
- Integrates OpenMS and DIA-NN
- Integrates three search engines: MSGF+, Sage and Comet (which can be all run for a search)

results/PXD040621

pipeline_info # parameters etc

pmultiqc # MultiQC report with file level quant information

quant_tables # peptide-level quantification

sdrf # Sample-Data-Relationship-Format (parsed and validated)

spectra # mzML files indexed (if raw files were parsed)

database: data/fasta/merged_ecoli_with_contaminants.fasta

input: data/PXD040621/PXD040621.sdrf.tsv

outdir: results/PXD040621

only relevant for 1.3.0

max_memory: 15 GB

max_cpus: 4

local_input_type: raw

local_input_type: mzML # default

! root_folder only applies to the mzML or raw spectrum files

root_folder: /workspaces/dsp_course_proteomics_intro/data/PXD040621/mzML/

publish_dir_mode: symlink

running msstats: Only two pdfs which we do not use.

skip_post_msstats: true

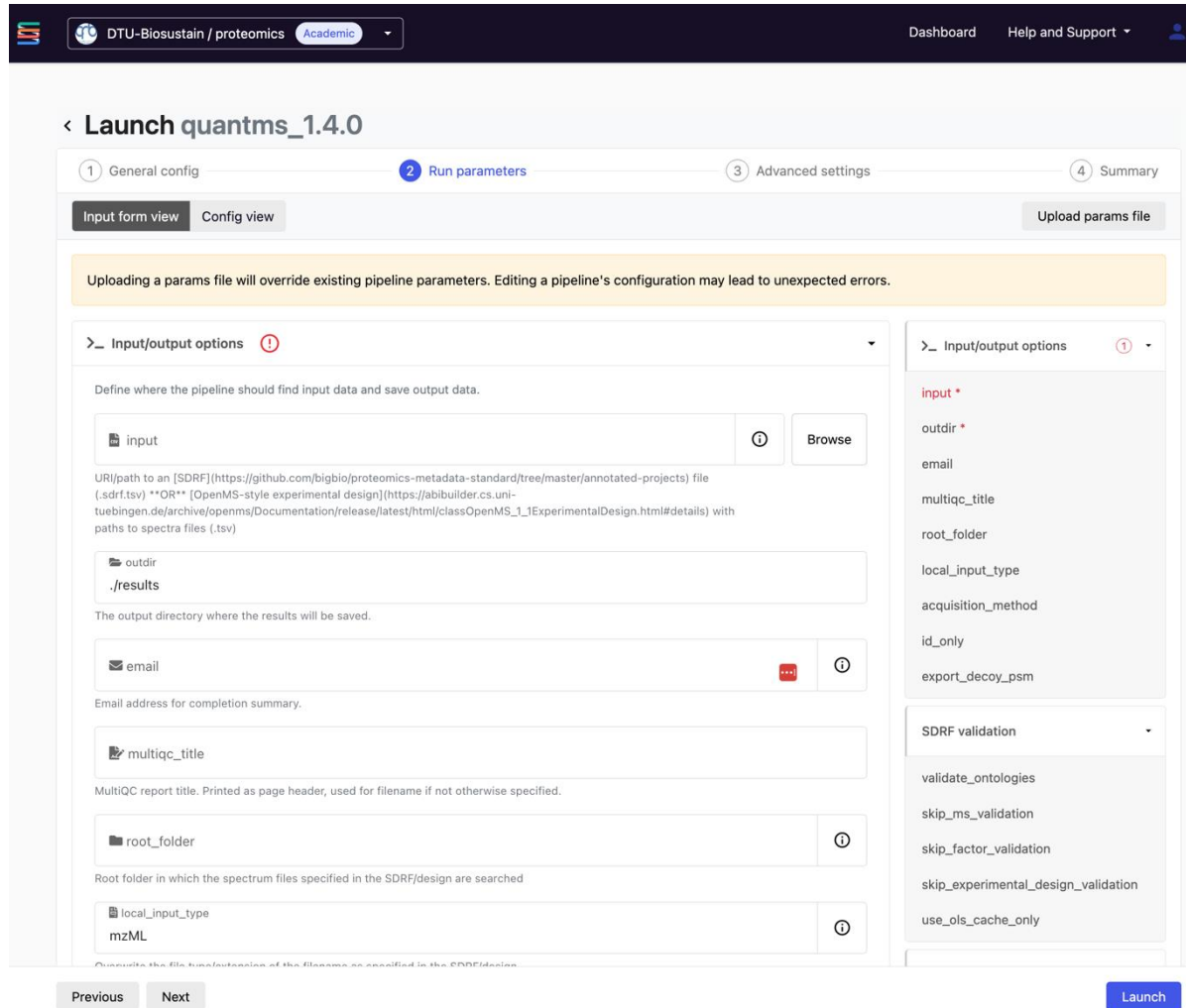
min_peptide_length: 6
max_peptide_length: 40

fdr_level: psm_level_fdrs
min_precursor_charge: 2
max_precursor_charge: 4

protein_quant: unique_peptides
min_peptides_per_protein: 1

precursor_mass_tolerance: 5
precursor_mass_tolerance_unit: ppm
fragment_mass_tolerance: 0.03
fragment_mass_tolerance_unit: Da
protein_level_fdr_cutoff: 0.01

Seqera interface for workflow



DTU-Biosustain / proteomics Academic

Dashboard Help and Support

< Launch quantms_1.4.0

1 General config 2 Run parameters 3 Advanced settings 4 Summary

Input form view Config view Upload params file

Uploading a params file will override existing pipeline parameters. Editing a pipeline's configuration may lead to unexpected errors.

> Input/output options ⓘ

Define where the pipeline should find input data and save output data.

input ⓘ Browse

URI/path to an [SDRF] (https://github.com/bigbio/proteomics-metadata-standard/tree/master/annotated-projects) file (.sdrf.tsv) **OR** [OpenMS-style experimental design] (https://abibuilder.cs.uni-tuebingen.de/archive/openms/Documentation/release/latest/html/classOpenMS_1_1ExperimentalDesign.html#details) with paths to spectra files (.tsv)

outdir
./results

The output directory where the results will be saved.

email ⓘ

Email address for completion summary.

multiqc_title

MultiQC report title. Printed as page header, used for filename if not otherwise specified.

root_folder ⓘ

Root folder in which the spectrum files specified in the SDRF/design are searched

local_input_type ⓘ

mzML

Configure the file type/extension of the filenames as specified in the SDRF/design

input *
outdir *
email
multiqc_title
root_folder
local_input_type
acquisition_method
id_only
export_decoy_psm

SDRF validation

validate_ontologies
skip_ms_validation
skip_factor_validation
skip_experimental_design_validation
use_ols_cache_only

Previous Next Launch

- Parameters grouped and in order
- Based on [schema file](#)
 - Schema file can be launched to see it website locally
- For more information: [Schema file documentation](#)

Data Analysis hands-on using acore

14th May 2025

Henry Webel

Aim: identify the impact of sulforaphane on the human gut microbiome (in anaerobic conditions of the gut)

Scientific Story:

- Phylogeny of selected strains and screening of growth kinetics of selected strains under sulforaphane in anaerobic conditions (Figure 1)
- Growth profile of selected *E. coli* Strain E2348/69 in anaerobic and aerobic conditions with varying amounts of sulforaphane
- Proteomics analysis of most growing strain
- Metabolomics analysis of most growing strain
- [Article](#) and [PXD040621](#) on Pride archive

“Proteomics identified an increase in anaerobic respiration in *E. coli* grown in the presence of sulforaphane, indicating suggesting that sulforaphane may be acting as an additional carbon source in these bacteria. The metabolic profile following growth in sulforaphane, showed that sulforaphane increased the production of metabolites that are also known to interact with host tissues to decrease inflammation.”

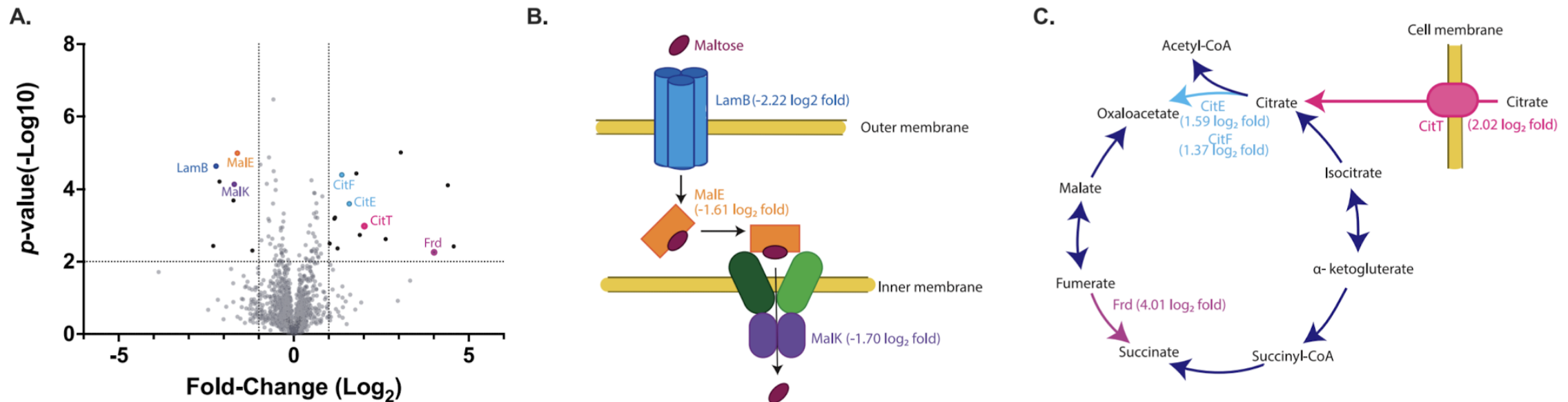


Fig. 3. Differentially produced proteins in *E. coli* EPEC ECE2348/69. **A.** Volcano plot of the change in protein production of individual proteins (dots) produced by *E. coli* ECE2348/69 in response to the presence of sulforaphane. Significantly altered proteins (black or coloured dots) showed a $\log_2$ fold-change < -1 or > 1 (vertical dashed lines) with a p -value < 0.05 (horizontal dashed lines). **B.** Schematic representation of proteins with significantly decreased production involved in maltose uptake. **C.** Schematic representation of proteins with significantly increased production (coloured labels) involved in anaerobic respiration.

MoNA and DSP Open Source Libraries



Analytics

Analytics core
library

analytics-core.readthedocs.io/



Visualisation



Visualization core
library

github.com/Multiomics-Analytics-Group/vuecore

Reporting



API

Web

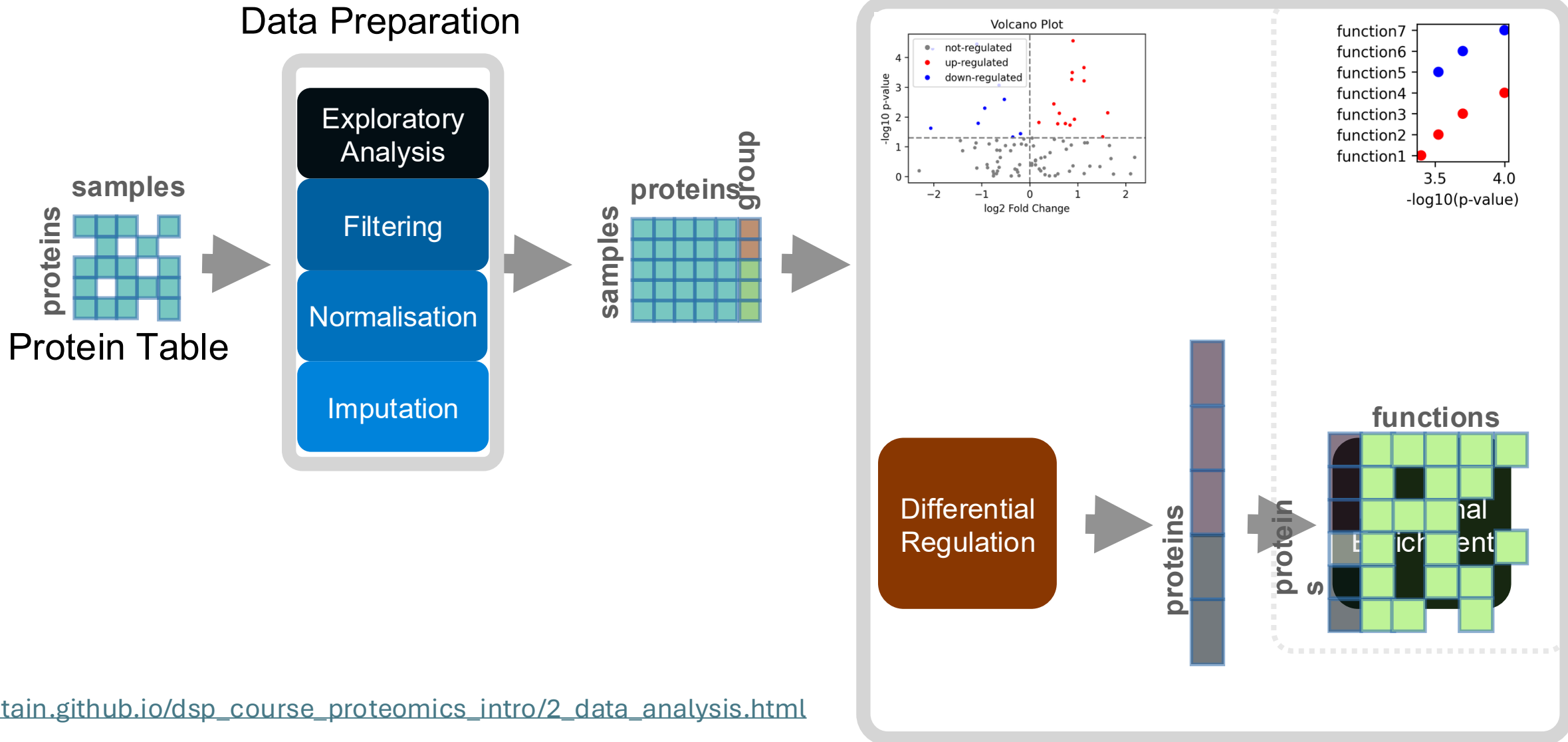
Doc

Automated Reporting
library and cli

github.com/Multiomics-Analytics-Group/vuegen



Acore – Analytical core – workflow example

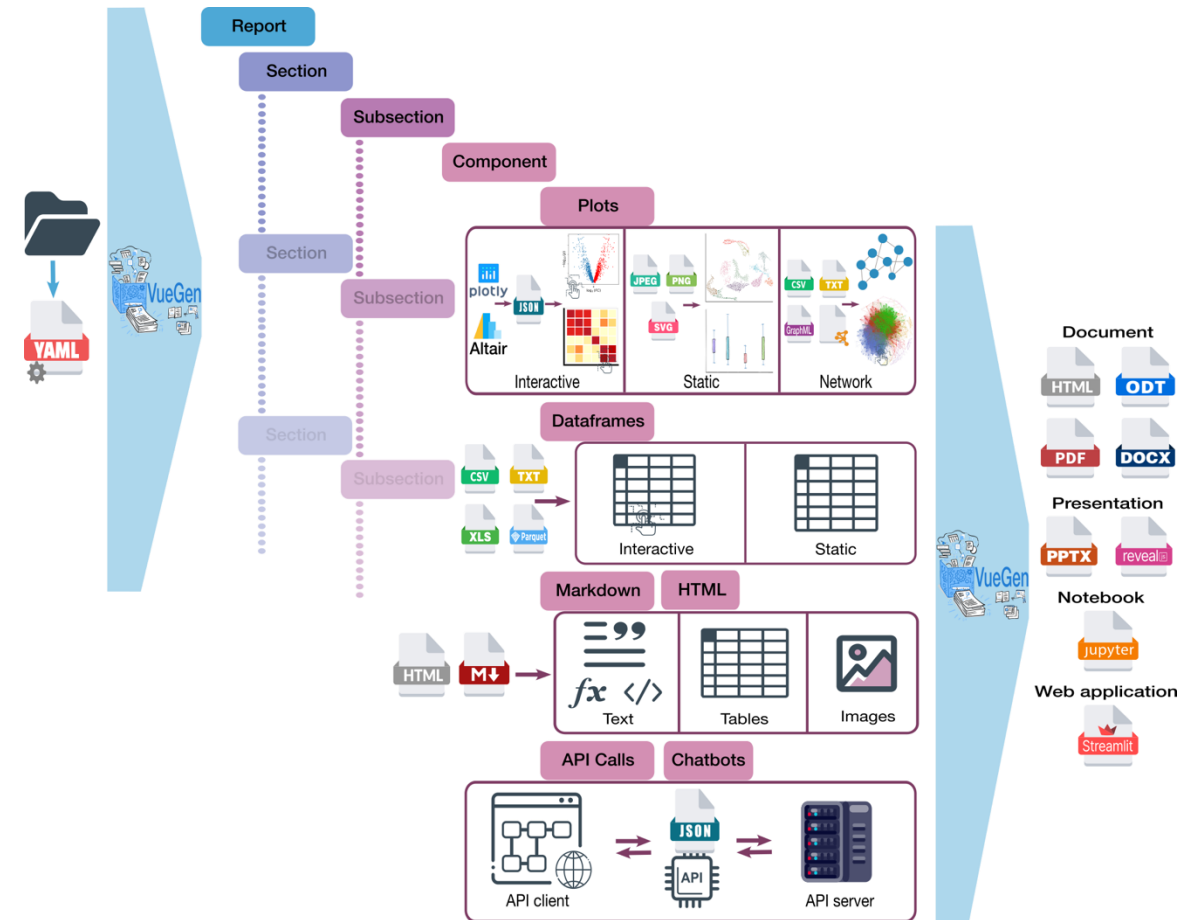


To Do

- Can be executed in an example notebook
- Which can be customized from the command line
- Which can operate on different datasets

Report Generation using Vuegen

- Turn results folder into different reports
- We collect plots and dataframes from our analysis example
- The analysis notebooks will export the relevant files



VueGen implementation details

